

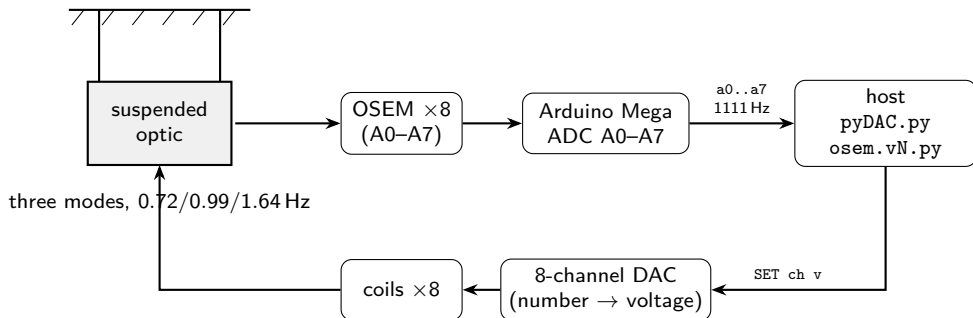
OSEMa bin-laden

Four rungs of a suspended-optic damping loop: zero, alpha, beta, delta

7 August 2026

Eight OSEM shadow sensors, eight coils, one suspended mirror,
and four kept rungs of the supervisor around them.

What is on the bench

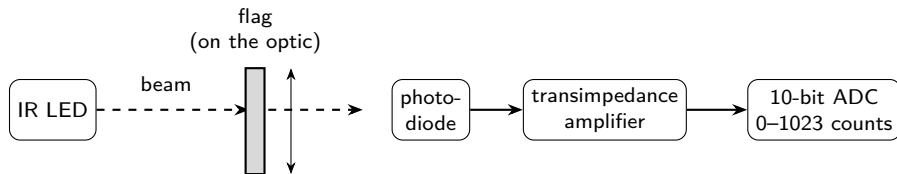


Sensor $i \rightarrow$ pin $A_i \rightarrow$ DAC_CHANNELS = [1,3,5,7,0,2,4,6].

All eight OSEMs and all eight coils are connected.

Four of them, a0-a3, carry usable in-band signal and their four coils push back.

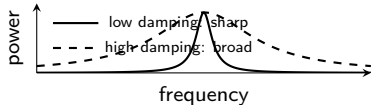
The OSEM: a shadow sensor



- ▶ **An analogue-to-digital converter turns that voltage into a whole number, 0 to 1023 “counts”.** Its two ends are the *rails*: past them the number cannot go further, whatever the mirror does.
- ▶ **At a rail the sensor is outside its linear region**, so anything derived from it is wrong until the flag comes back.

How the damping was measured, and why it is a bound

A 320 s dead read w/ coils parked, no kick: 355 797 samples.



Why that still gives the damping. Shaken by broadband noise, a resonance makes a peak, and the **width** of the peak is set by the damping: $\text{width} = f_0/Q$. Measure the *sharpness of the peak*, not the decay of a swing.

276 s of usable record resolves $1/276 = \mathbf{0.0036\text{ Hz}}$
 $\mathbf{0.00543}$.

The plant: three resonances, and a bound on the damping

A suspended optic is a damped harmonic oscillator. For one sensed coordinate $x(t)$, mass m , force $F(t)$, and mode m :

$$\ddot{x} + \frac{\omega_m}{Q_m} \dot{x} + \omega_m^2 x = \frac{F(t)}{m}$$

Amplitude decays as $e^{-\gamma_m t}$, so

$$\gamma_m = \frac{\omega_m}{2Q_m}, \quad \tau_m \equiv 1/\gamma_m.$$

Resonances, 320 s ringdown, coils at bias:

$$f_A, f_B, f_C = 0.7155, 0.9949, 1.6396 \text{ Hz}$$

$$\pm 0.0001 \text{ Hz}$$

γ_m across channels, in s^{-1} :

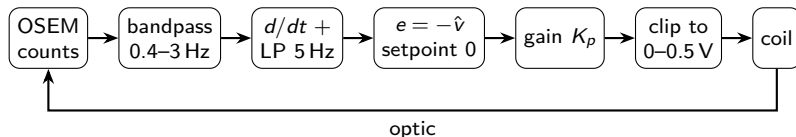
	ch	median	range
A	4	0.0483	0.0107–0.1388
B	5	0.0082	0.0005–0.0579
C	4	0.0094	0.0061–0.0530

Cold damping: the process variable is *velocity*

$$\text{bp}(t) = \text{LP}_{3\text{ Hz}} \left[\text{HP}_{0.4\text{ Hz}}[A(t)] \right]$$

$$\hat{v}(t) = \text{LP}_{5\text{ Hz}} \left[\frac{\text{bp}(t) - \text{bp}(t - \Delta t)}{\Delta t} \right]$$

$$u(t) = -K_p(t) \cdot \hat{v}(t)$$



All three measured resonances sit inside bandpass window. The target is a constant zero, so the error is just minus the speed: the loop never asks where the mirror *is*, only that it stop *moving*.

Three gains: P is a damper, I is a spring, D is a mass

A **PID controller** combines three responses to an error: one proportional to it, one to its running total, one to its rate of change. Here the “error” is the mirror’s speed, which shifts what each term means.

$$\underbrace{K_p e = -K_p \dot{x}}_{\text{force} \propto \text{velocity}}$$

$$\underbrace{K_i \int e dt = -K_i x}_{\text{force} \propto \text{displacement}}$$

$$\underbrace{K_d \frac{de}{dt} = -K_d \ddot{x}}_{\text{force} \propto \text{acceleration}}$$

term	acts on	physically	effect on the pole
K_p	velocity	viscous damper	the only term that removes energy
K_i	displacement	added spring	moves the resonance; does not damp
K_d	acceleration	added (negative) mass	noisiest; differentiates twice

Why add I and D at all? K_p sets the damping, but it leaves a steady offset it cannot remove and it reacts only to present error. K_i takes out the offset; K_d anticipates (and therefore useless for us)

How hard to push: choosing the gain

1. **Two ceilings stop it going up indefinitely.** Above about 0.040 the loop was seen on an oscilloscope to go unstable. And the coil drive has a fixed $\pm 0.25\text{ V}$ window, past which the loop asks for more push than the hardware can give.
2. **0.035 fits under both, and *instability* binds:** the budget at 0.035 is **0.2001 V** of 0.250 V, so the hardware still has room when stability runs out.

	ch0	ch1	ch2	ch3	ch4-7
K_p	-.035	-.035	+.035	-.035	0
K_i	-.0060	-.0060	+.0060	-.0060	0
K_d	-.00045	-.00045	+.00045	-.00045	0

ch2 is positive, and that is correct. Its sensor or coil are mounted the opposite way round.

Only $|K_p| = 0.030$ was fixed by watching an oscilloscope: 6 $\times$ reduction, no clipping.

The gain is not one fixed number

Three definitions, used for the rest of the deck.

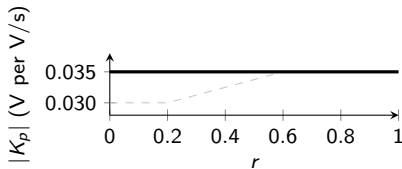
- ▶ **Envelope:** the running size of recent motion, an RMS of the bandpassed sensor signal. Window per job, 1 / 2 / 5 s (`osem.delta.py`).
- ▶ **Baseline:** the envelope measured once per run with the gain forced to zero. This room, this night, nothing driving.
- ▶ r = envelope now / that channel's own baseline. $r = 1$ is "as lively as when we were not pushing".

Why declare two gain levels at all. K_p is the strength of the artificial drag: volts onto the coil per volt-per-second of sensor speed. Large motion wants the stiffest drag the hardware allows, to be caught quickly. A mirror near rest does not: its sensor signal is mostly noise, so a stiff drag pumps that noise back into the coils, and the coil drive has the fixed ± 0.25 V window from the previous slide. Hence a **capture** level for large r , a **steady** level for small.

That is the argument in general. **No justification for *this* rig having two levels is recorded anywhere in the project:** not in the source, not in `versions.md`, not in a docstring.

One number, measured once, in one window, decides all three of these.

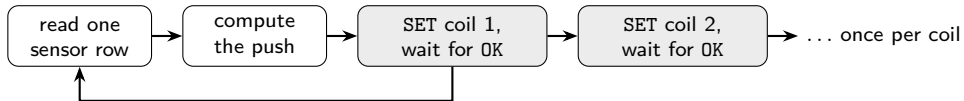
- ▶ **The fault threshold.** The runaway breaker trips above $1.8\times$ baseline.
- ▶ **The gain blend.** Full capture above $r = 0.6$, tapering to steady by $r = 0.2$.
- ▶ **The definition of success.** LOCKED means below $0.35\times$ baseline, held 5 s.



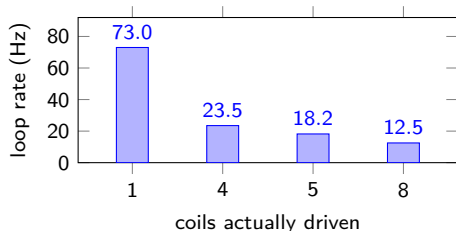
Solid: what ships. Both levels are ± 0.035 , so the schedule is flat and the blend is inert. Grey dashed: the shape the code can express, never used. Still live: the rate limit, 0.02 per second, and the soft-start it implies, $0.035/0.02 = 1.75$ s from zero.

The cause: one coil write is one round trip down the wire

What the host does to move a coil. It sends one SET ch v line, then waits for the board to answer OK. While it waits, it reads and discards up to **50** lines of the sensor stream.



Grey blocks are the wait. **The loop rate is set by how many coils are driven, not by the control law.**



Median DAMPING-row spacing **13.7 / 42.6 / 55.0 / 79.9 ms**. The 1, 4 and 8-coil points are medians over **20 bench logs** (analysis/out/loop_rate.csv); the 5-coil point is the 2026-08-06 run (versions.md).

The strongest form of the claim is one run. 357.1 Hz while CALIBRATING, where no coil is written, against **18.2 Hz** while DAMPING on 5 coils. Same run, same minute, **a factor of 19.6**, so there is no between-run variation to argue about.

Two consequences of a loop rate that moves

1. One threshold was counted in samples, not seconds.

MAX_CONSECUTIVE SATURATED = 30 trips after 30 consecutive samples with the coil command pinned at the end of its window. It is the only threshold in the codebase counted that way, so its real meaning moved with the coil count:

coils driven	1	4	5	8
loop rate (Hz)	73.0	23.5	18.2	12.5
30 samples (s)	0.41	1.28	1.65	2.40

Bottom row: what 30 consecutive samples amounts to in wall-clock seconds.

The interlock got **less sensitive exactly as the rig got harder to control**: same 30 samples, six times longer on the wire.

2. Every gain in the rig was fixed on a 4-coil loop.

A feedback loop always pushes a little late.

Phase margin is how much more lateness it can absorb before the push lands *with* the motion instead of against it, at which point the loop adds energy rather than removing it. Quoted in degrees of one cycle, at a stated frequency.

At **3.75 Hz**: **73.5°** on 4 coils, **45.5°** on 8 (analysis/out/phase_budget.csv). Carrying the same gains onto eight coils spends **28°** of a budget nobody re-measured.

Read 3.75 Hz with care. It is one of the fixed evaluation frequencies in that file, not one of the three measured modes (0.7155 / 0.9949 / 1.6396 Hz), and **nothing in the project records why it was chosen**. Take the pair as a comparison of two configurations, not as a margin at a frequency that matters.

zero: the original

ENABLE_CHANNEL = [True, False, False, False]
P-only: $u = -K \cdot v$, KI_GAIN = KD_GAIN = 0.

What zero got right, and every version since inherits:

1. Runs continuously, not as fixed-duration sweep steps.
2. Measures its *own* baseline every restart.
3. A rail interlock on *raw counts*, separate from the RMS breaker. A blind OSEM flatlines, which an RMS-only check reads as perfect stability.
4. Auto-recovery out of FAULT.
5. A lock detector that prints a real number.
6. Streaming CSV, flushed periodically, survives an unclean kill.
7. Gain scheduling with a slew-limited ramp.

zero, through harness.py

ch0 amplitude ratio	0.188
time to lock	12.1 s
suite (when present)	18/18
undriven ch1/2/3	0.333 / 0.316 / 0.253

Three defects shipped with it and survived into the next two builds: saturation-latch, rail-threshold, runaway-baseline. All three are fixed in alpha.

Plus a fourth nobody found until beta, and a fifth until delta.

Defects fixed without argument: the three zero shipped

An **interlock** is an automatic check that pulls the loop out of DAMPING and parks the coils at bias; it **arms** once it has enough evidence to fire. The **runaway breaker** is the interlock watching for the loop feeding the motion instead of removing it.

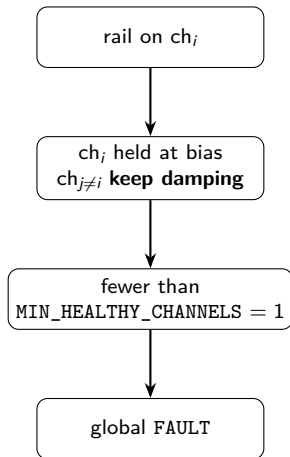
All three below were found in zero and answered in alpha. Each gets one line, because stating it is the same as explaining it.

1. **saturation-latch**. *Intended*: count consecutive samples with the coil command pinned at a rail, fault above 30. *Actual*: the flag was recomputed only inside `check_runaway()`, which does not run in FAULT, so once it tripped nothing cleared it: a permanent fault in a module advertising auto-recovery. *Evidence*: across **16 966** FAULT samples the run logged **one distinct amplitude ratio, 3.6803**. It latched ~ 50 s. *Fix*: refresh the flag every sample in every state; in FAULT the output sits at bias, so it clears itself. alpha, same bench: **0 pinned samples, 0.00% ADC clipped, no fault**.

2. **rail-threshold**. *Intended*: call a channel blind when its raw counts stay under `RAIL_LOW_COUNTS = 3` of 1023 (14.7 mV) for 250 consecutive samples. *Actual*: a blind channel still carries about one count of dark noise, so one sample above 3 resets the counter and the interlock never arms. *Evidence*: a four-channel bench run logged **246** genuinely clipped ADC samples, **2.19%** of the run, and **the flag never set once**. *Fix*: threshold $3 \rightarrow 12$ counts, and the test becomes 80% of a 0.5 s window rather than an unbroken run. Arms in **0.41 s** at 0, 5, 10 and 20 mV of injected noise; the old test **never armed** at 10 or 20 mV.

3. **runaway-baseline**. *Intended*: one RMS over one 8 s window at zero gain gives the baseline. *Actual*: a bump inside that window inflates the baseline, and every threshold derived from it, for the whole run. *Evidence*: three runs, same rig, same afternoon, ch0 baseline **0.6316 / 0.3158 / 0.1804 V**, a **3.5 $\times$** spread, while the absolute locked RMS was **0.139 / 0.128 / 0.115**. Equally well damped; only the yardstick moved. *Fix, partial*: calibrate 20 s, take the median of 5 sub-windows. Worst shock skew 39.3% $\rightarrow$ **19.7%** against a 35% assertion line (simulator, 6 V/s impulse swept across the window). **The code calls it partial**; defect 6 finishes it.

alpha: park one blind sensor, do not freeze the rig



The problem. alpha freezes all four channels on any one rail. One OSEM losing its flag stops *all* damping, at the moment of largest excursion, which is what made the flag leave range.

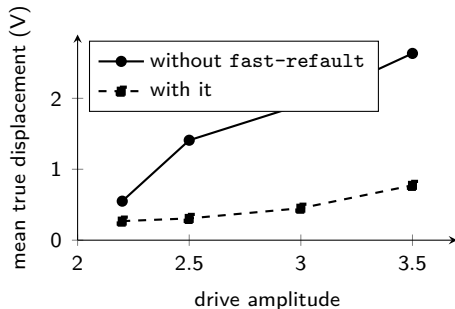
A quorum of one is not a guess: the zero campaign measured one OSEM/coil pair damping the whole rigid body.

condition	alpha	why
rail	demote	local <i>sensing</i> failure: that measurement is garbage
runaway	global	not localisable: four loops, one body
saturation	global	measurement fine, only the output clipped

Saturation is deliberately *not* demoted: a saturated channel is still pushing the right way at full authority.

First fired on hardware at the eight-channel step: ch2 railed, 3 of 4 kept damping, re-armed 2 s later.

beta: stop re-measuring during the event



Simulator *ground truth* over 300s, not the controller's own estimate.

The problem. Every build before this one answers every fault the same way: wait for all-clear, then recalibrate. That is CALIBRATION_S of open loop at zero gain. Correct for a transient, wrong for a disturbance that has not gone away: the optic rings straight back up to the amplitude that faulted it.

200 s @ drive 2.5	% DAMP	faults	locks
earliest builds	0.6%	1	never
with the 3 fixes	4.1%	7	never
with fast-refault	87.5%	1	10.2 s

The fix is four lines. If the previous engagement lasted under FAST_REFAULT_S = 15s, the cause is still present, so re-engage on the baseline already in hand. Bounded by MAX_BASELINE_REUSE = 4.

Found by stressing the simulator across clipping regimes, not by reading the code.

Defect 1: a silent sensor takes the whole rig down

1. What it was supposed to do. Each channel measures its own baseline, then faults if its envelope climbs past $1.8\times$ that number. A per-channel yardstick, so quiet channels are held to quiet standards.

2. What it did. A channel carrying no signal at all calibrates a baseline of almost nothing, so its own electronics noise clears $1.8\times$ easily and it faults constantly. Runaway is a **global** fault, because “amplitude is growing under feedback” does not say which of four loops is feeding it. So one silent sensor stops all eight.

3. The evidence. Calibrated baselines when eight channels were first enabled (2026-08-04):

	a0	a4	a5	a6	a7
baseline (V)	0.3109	0.0014	0.0801	0.0050	0.0020
$1.8\times$ trip line	0.560	0.0025	0.144	0.0090	0.0036

a4's bandpassed output never left ± 0.01 V, which is $4\times$ its own 0.0025 V trip line.

Measured peak ratios: a4 **3.98**, a6 **4.79**, a7 **11.31**, all on a signal too small to see. Cost on the bench: **ten whole-rig faults in 145 s, nine of them naming a4, a6 or a7.**

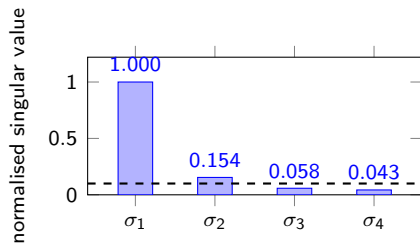
Why no existing interlock caught it. An OSEM with no in-band signal does not reach a rail. It sits mid-scale with a flat bandpassed output, which every RMS check reads as *perfect stability*, so auto-disable never fires.

4. The fix, and the number that moved.

baseline-floor, now in delta: at the end of every calibration, demote any channel whose baseline is under **10% of the median** across enabled channels.

2026-08-06, same eight OSEMs: a4/a6/a7 demoted at $t = 20.001$ s, then **40 s of unbroken DAMPING and zero faults.**

sysid: measuring which coil moves what



Dashed: the 10% retention line. **Directions above**
it: 2

KIND = "sysid". **Not a controller**: it damps nothing and closes no loop.

Stepped sine, one coil at a time. Drive one coil at a single frequency and measure only the response *at that same frequency*, which rejects everything else. While coil j drives, no other coil is, so there is nothing to disentangle and **both the size and the timing of the response fall straight out.**

Only **4 of 12** frequencies were usable: ch2 railed in **17 of 48** steps, and two survivors are ambient-dominated.

A multisine variant failed outright: ch2 railed in **all 4** passes, and pass 1 ran at **14.8 Hz** effective against ~ 252 Hz, giving a spurious ~ 4700 V/V against ~ 70 . **One cause for both**: open loop, nothing damping, ch2 clips. **And the rank check is superseded**: the ringdown finds **three** modes, and the full refit puts the allocator at 3×4 , cond **1.41**.

The DC actuation matrix, and the conclusion it will not support

	coils 0–3				coils 4–7			
a0	−105	−107	−48	+19	−29	−21	−28	−26
a1	−49	−68	−33	+18	+5	+7	+4	+8
a2	+35	+66	+213	−141	−25	−22	−13	−26
a3	+36	+14	+58	−51	−1	−2	+2	−1
a4	+0	+2	+2	+1	+8	+8	+2	+1
a5	+4	+0	+0	−6	−0	−7	−1	+8
a6	+1	+4	+2	+3	+3	+3	+4	+6
a7	−0	+3	+1	+0	+1	−1	+6	+7

$d(\text{counts}_i)/d(\text{bias}_j)$, one coil stepped at a time, bench

2026-08-04.

	Block means of response	
	coils 0–3	coils 4–7
sensors 0–3	66	14
sensors 4–7	2	4

Coils 4–7 work: they move a0–a3 by ~ 25 counts/V. **Sensors a4–a7 respond to nothing,** ≤ 8 counts/V.

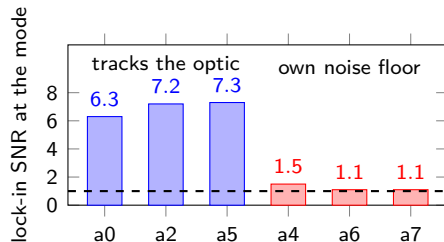
We read that as “a4–a7 are not wired”. It was wrong. A DC slope against a fixed threshold cannot separate *no sensor* from *a working sensor with small gain*. **Wrong statistic, not wrong data.**

Their own diagonals are +8/ +4/ +7 against a block mean of **4**. Those signs are coin flips, and **a wrong sign pumps**, so delta ships a4/a6/a7 at zero gain.

a3: the common-mode sweep read +43 counts/V, coil3 $\rightarrow$ a3 alone −51.

Common-mode slopes are not per-coil slopes.

All eight *are* wired: in-band against out-of-band



Lock-in on the two long 8-channel logs (118s, 148s)
at the mode frequencies. Dashed: SNR = 1.

All eight OSEMs are electrically connected, confirmed on a scope. The split is *in band* against *out of band*:

power in 0.4–3 Hz		
a0–a3	77–99%	the three modes
a5	22–71%	in band here, not in the residue fit
a4, a6, a7	0.1–9%	a narrow 6.19 Hz line (12.4 Hz harmonic on a7)

Not an Arduino or sampling problem. a3 is read *immediately before* a4 and carries a large 1.64 Hz signal, yet $r(a4, a3) = -0.029 / +0.004$. a5, read straight *after* a4, is clean.

The cause of the low gain is NOT established. “Flags outside the linear region” is **unsupported**: a4 rests at **567.5** counts against mid-scale **511.5** with a **30-count swing**, nowhere near an end.

Defect 2: the runaway breaker was asking the wrong question

1. **What it was supposed to do.** Stop the loop when it is *feeding* the motion. A **runaway** means a **growing** envelope; a **ringdown** is a large but **falling** envelope, the loop succeeding.
2. **What it did.** It only asked whether the envelope was *large*, never whether it was growing, and a level test cannot tell those apart. So after a kick it faults on a timer, every RUNAWAY_SUSTAIN_S, all the way down a ringdown it should have left alone.
3. **The evidence. The build just before beta: hardware, 240 s, three kicks, ten faults, envelope falling through them** (next slide). The defect was in every version back to zero; that build merely re-engaged fast enough after a fault to hit it repeatedly, which is how it surfaced.
4. **The fix, and the number that moved.** beta keeps the level test and **ands** a growth test onto it. Both conditions, both sustained.

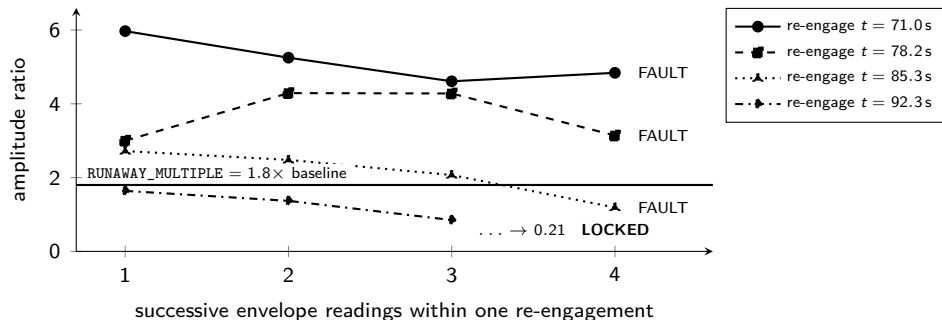
level test (every build before beta): $\text{env} > 1.8 \times \text{baseline}$, sustained 2 s

growth test (new in beta): $\text{env} > 1.02 \times \text{env}(t - \text{LAG})$

both $\Rightarrow$ FAULT

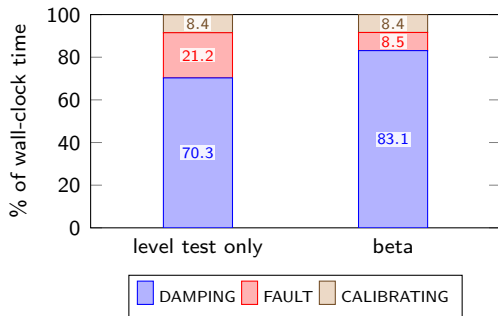
Same 240 s three-kick schedule, growth test off then on: **10 faults $\rightarrow$ 4**, DAMPING **70.3% $\rightarrow$ 83.1%** of wall-clock time. The 1.02 is 2% of headroom, so envelope noise alone cannot read as growth. But RUNAWAY_TREND_LAG_S = 2.0 s was picked before any mode frequency had been measured. **That lag is defect 5.**

The defect on the bench: the envelope *falling* through every fault



Measured on **the build just before beta: hardware, 240 s, three kicks: ten faults**, and the envelope was *falling* through them. Only the re-engagement that happened to start at 1.64, just under the line, survived.

beta: what the growth test bought



The same build with and without the growth test, on the same 240 s three-kick schedule.

same 3-kick schedule	faults	locks
level test only	10	4
beta	4	4

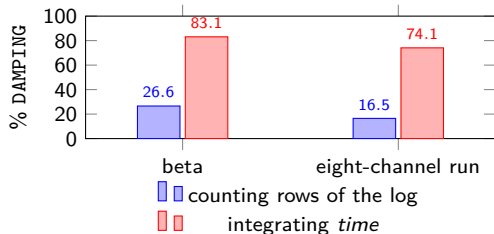
beta re-locked after *every* kick, at $t = 31.3/83.8/151.3/205.3$ s.

beta's FAULT time is **20.2 s** = the mandatory 5 s all-clear $\times$ 4 recoveries. **Cost, not thrash.**

83.1% is still the best DAMPING fraction on record. Every later comparison has to say what kick schedule it ran.

Defect 3: two estimators, one inverted conclusion

1. What it was supposed to do. Report what fraction of a run the loop spent damping, so two builds can be compared. **2. What it did.** Two estimators of that fraction were in use and one number from each was divided by the other: **rows** counts samples in the CSV, **time** integrates the timestamps.



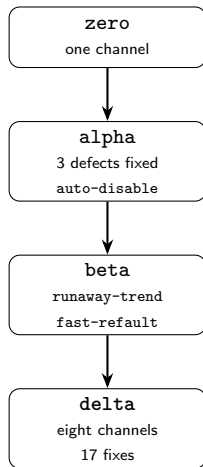
Identical logs and state column. Only the weighting differs.

3. The evidence. The eight-channel run's **74.1%** (time) was set against beta's **26.6%** (rows), reading as **a 2.8× improvement that does not exist**. Like for like, either way, beta wins: time **83.1** vs 74.1, rows **26.6** vs 16.5.

Why rows lie. The loop samples at **24–26 Hz** while actuating and **341–351 Hz** otherwise, because only actuating pays the serial round trip, so counting rows weights each state by how fast it happens to sample.

4. The fix, and the number that moved. Integrate timestamps. The distortion is no constant to correct for: rows understate DAMPING by 3.1× on beta and 4.5× on the other run, so row-based figures are not comparable to *each other* either.

delta: a clipped loop is weakened, not broken



Four kept rungs. Further builds ran between them and are in version control only.

The fix. Earlier builds faulted the whole rig after 30 consecutive samples with an output pinned at its limit. But **clipping removes push in one direction only**: an output stuck at the top of its range can still pull all the way down at full strength. A clipped loop is a **weakened** loop, not a broken one, and it is usually the thing bringing the mirror back.

$$\text{sat_trip} = (\text{streak} > 30) \wedge \neg(\text{amplitude falling})$$

The same level-against-trend correction the runaway breaker got, one interlock over. Same history, same lag, so “falling” means one thing to both.

And a coupling worth knowing. Nudging a coil's resting current is a force step: it rings the pendulum inside the band both trend tests read. So a nudge **invalidates** the history instead of suspending the interlocks, which would open a 16 s hole every 15 s.

bias-trim inside a live loop

trim step	total offset	verdict
ch3 → 0.50 V	576 → 560	kept
ch3 → 0.75 V	560 → 532	kept
ch3 → 1.00 V	532 → 507	kept
ch1 → 0.50 V	507 → 472	kept
ch3 → 1.25 V	472 → 564	reverted
ch1 → 0.75 V	489 → 477	kept
ch1 → 1.00 V	477 → 481	reverted
ch0 → 0.50 V	501 → 522	reverted

Total |offset from mid-scale|, four OSEMs, counts.

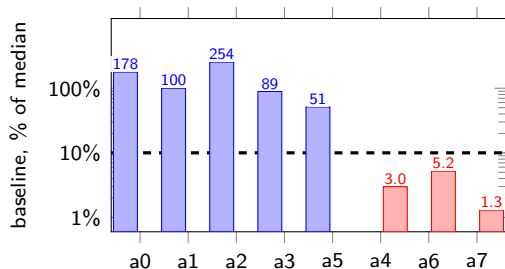
Accept-or-revert earns its keep. 3 of 8 steps made the *rig* worse and were frozen. ch3 at 1.25 V cost **92 counts** in one step: four coils, two DOF.

More faults, no thrash. 35.7 s of FAULT over 7 trips is **5.1 s each**, the mandatory FAULT_CLEAR_SUSTAIN_S, as beta's 5.05 s.

ch3 caused 7 of the 8 events (ch1 and ch2 one each). It has the **weakest coil** at −51 counts/V against ch0's −105 and ch2's +213, and it is the channel the trim stepped three times. **A bias step is a force step.** *Trim the weakest coil last.*

soft-saturation never fired: every trip was the runaway breaker. But auto-disable did, for the first time on hardware: ch2 railed, **3 of 4** kept damping, re-armed 2 s later.

baseline-floor: the guard eight channels needed



Dashed: $\text{BASELINE_FLOOR_FRAC} = 10\%$ of the median over *enabled* channels, 2026-08-06. **It sits in an empty**
 $10\times$ **gap**: 5.2% (worst dead) to 51.3% (a5).

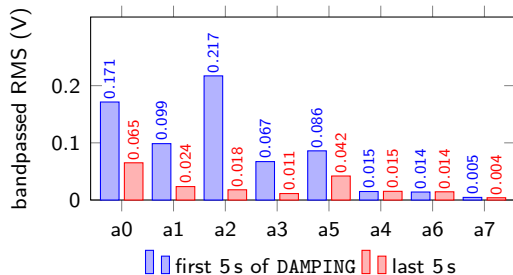
The rule. At the end of every calibration, a channel whose baseline is under 10% of the **median** across enabled channels is demoted exactly as auto-disable demotes a railed one.

Median, not mean, and that is the whole safety argument. The largest baseline is always $\geq$ the median $>$ the threshold, so **at least one channel always survives**: no quorum arithmetic. And if *most* of the rig is dead the median *is* a dead channel, the threshold collapses, and the guard does nothing, rather than demoting six on the word of two.

Relative, not absolute, because an absolute floor in volts would depend on the seismic background and each OSEM's alignment. On a quieter night a real channel and a dead one fall together.

And note what a config switch did *not* do. Those three were switched off in the configuration and still had to be demoted: the runaway breaker gates on healthy, not on enabled. Switching a channel off does not take it out of the interlocks.

a5: a sign predicted out of noise, and then retracted



40 s of damping, eight channels.

`STEADY_GAIN[5] = -0.015` was a **prediction**: a5's DC diagonal is -7 counts/V against a block mean of **4**, under $2\times$ the noise. A wrong sign *pumps*.

It damped: $0.0860 \rightarrow 0.0420$ V, while ch0 fell $0.1714 \rightarrow 0.0651$.

The three at zero gain are the control.

a4/a6/a7 are **flat to three decimals** across 40 s that took a factor of **3–12** out of every real channel.

Then the newer measurement retracted it. The fixed-frequency refit (2026-08-06) finds a5's residue **not distinguishable from its own noise**, and a gain on a channel that measures noise *injects* actuator noise while damping nothing.

delta **ships** `STEADY_GAIN[5] = 0`. The two measurements disagree and **the one with error bars wins**. a5 is still in the quorum while nothing drives it, which is what holds the *eight-channel* configuration out of LOCKED.

delta: eight channels, and the wire stops throttling the loop

1. sat-window: a bug fix, not a retune.

MAX_CONSECUTIVE_SATURATED = 30 counted *samples*, so it meant 0.41 / 1.28 / 1.65 / **2.40**s at 1 / 4 / 5 / 8 coils, and one clean sample erased the whole streak.

Now SAT_SUSTAIN_S = 1.30 s with SAT_FRACTION = 0.80: the shape the *rail* check has always had. $30 \times 42.6 \text{ ms} = \mathbf{1.278 \text{ s}}$, so it is **behaviour-preserving on four channels**.

It goes **first**, because the other two changes both move the sample rate. *You do not move a sample rate underneath a sample-counted threshold.*

2. fast-transport. pyDAC2.FastDAC writes and returns with no ack wait, and the baud goes 115200 $\rightarrow$ 500000. **18 Hz $\rightarrow$ 1111 Hz.** Requires a reflash; a host at the wrong baud fails preflight with “never sent READY”.

3. All eight enabled, with a4/a6/a7 as *sensors* at **zero gain**.

baseline-floor demotes all three on the first calibration, and that is the designed outcome. ENABLE_CHANNEL stops being a constant somebody maintains and becomes a **measurement the rig makes every time it calibrates.**

And then it damped worse than everything before it.

240 s, 3 kicks	DAMP	FAULT	faults
beta, 4ch, 23 Hz	83.1%	8.5%	4
8ch, 18 Hz	74.1%	15.0%	8
8ch, 1111 Hz	42.3%	25.3%	11

A 49 $\times$ faster wire made the damping worse.

Defect 4: a torn serial frame, and 45 V into a 0.5 V rail

1. What it was supposed to do. Reject anything on the wire that is not a sensor row. A **torn frame** is a line that is not one complete row: two partial writes spliced together at a buffer boundary. **2.**

What it did. `read_sample()` checked the row's *shape*, not its *values*: any line starting with a digit and splitting into $\geq N$ fields was accepted, and a torn line passes that test.

valid, every field in 0..1023

612,634,701,688,559,548,861,733

torn: 8 fields, first char a digit, accepted

612,634,522676,688,559,548,861,733

Illustrative numbers, not measured data. Row shapes are drawn by hand; **522676** is a real value from the log.

3. The evidence. fast-transport stopped draining the board's OK `ch=.. v=..` acknowledgement, so acks interleave with data rows. That build's own 240 s log carries **ten** rows with a count outside 0..1023 (5659, 65690, 522676). **Zero such rows appear in any of the four slower logs from the same session.**

	value	step
1	522676 counts	the torn field
2	$\times 5.02/1023$	= 2564.8 V raw
3	$\times \alpha_{LP} = 0.0178$	= + 45.6 V into bp
4	rings down, $\tau_{HP} = 0.398$ s	the 0.4 Hz highpass
5	$\hat{v}$ peaks	= 1401 V/s
6	$P = 0.030 \times 1401$	= 42 V demanded
rail		0.5 V

Worked through the shipping filter chain from one real torn row, not a logged measurement.

One bad character pins the actuator for the whole 0.4 s ringdown of the highpass. **Four of that run's twelve entries to FAULT follow a torn line within 2 s**, and the first starts the thrash that eats the middle of the run. **The fix is the next slide.**

sample-guard and decimate, and why they ship together

The fix, and the number that moved.

sample-guard: a count outside

0..ADC_MAX_COUNTS is a **framing error**, dropped exactly the way an OK line is. Three lines.

Replayed offline over the same log, ch0's velocity estimate:

ch0 velocity	before	+guard
rms (V/s)	13.109	5.413
peak (V/s)	1441	41

A 35× reduction in peak, on data already recorded. No bench time, no hardware change.

decimate: give the loop its own clock. Read the wire at full rate; run the control step at $\text{CONTROL_HZ} = 100$ on the *mean* of whatever arrived since the last one.

Why: the arrival rate has been 24/12.5/18/1111 Hz, and every gain, filter and interlock here was tuned against one of them.

ch0 velocity rms (V/s)	with guard	without
before	13.109	13.109
+ decimate	4.732	23.4

Without the guard, decimation makes it worse.

A boxcar mean *spreads* one bad sample across a whole step instead of confining it to one. **The two are not separable.**

Phase cost is measured: 100 Hz gives back 13.1° at 3.75 Hz against the raw 1111 Hz wire, and still holds 18.4° *more* than the four-coil configuration every gain here was validated at.

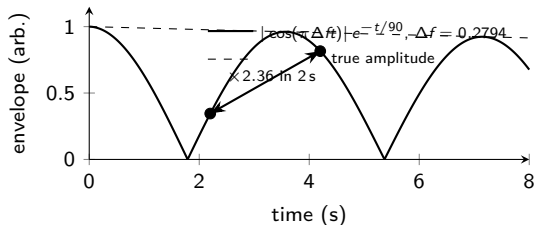
Defect 5: the runaway breaker was reading the *beat*

- 1. What it was supposed to do.** beta's growth test compares the envelope now against the envelope one lag ago, and faults if it grew by more than 2%. The lag, $\text{RUNAWAY_TREND_LAG_S} = 2.0\text{ s}$, was chosen before any mode frequency had been measured.
- 2. What it did.** Three modes sounding together make the envelope *beat*: it rises and falls at the difference of their frequencies while the true amplitude only decays. A 2 s lag lands the two readings on different phases of that beat, so the test measures the beat and calls it growth.
- 3. The evidence.** Modes measured to **0.0005 Hz** on the 320 s quiet record. Beat period = $1/|f_i - f_j|$:

pair	$ \Delta f $ (Hz)	beat period
A-B	0.2794	3.578 s
B-C	0.6447	1.551 s
A-C	0.9241	1.082 s

$$f_A = 0.7155, f_B = 0.9949, f_C = 1.6396\text{ Hz.}$$

The 2 s lag sits inside all three beat periods.



Illustrative numbers, not measured data. Two equal sines at the measured A-B spacing, times a 90 s decay. At $t = 2.2\text{ s}$ the envelope reads **0.3455**; at $t = 4.2\text{ s}$ it reads **0.8154**. The test asks for $> 1.02\times$ and gets **2.36 $\times$** , while the *true* amplitude has *fallen 2.2%*. The fix is on the next slide.

Defect 5, other half: a real ringdown was invisible too

2. What it did., second way. τ_{env} , the decay constant of the envelope the breaker watches, measures **56–130 s** after a kick.

	$\tau_{\text{env}} = 56 \text{ s}$	130 s
fall in 2 s	3.5%	1.5%
fall in 10 s	16.4%	7.4%
threshold: 2%		

3. The evidence. At $\text{LAG} = 2 \text{ s}$ a genuine ringdown moves **1.5–3.6%**, straddling the 2% line. **Broken in both directions:** a beat read as growth, a real decay was barely visible.

4. The fix, and the number that moved.

`RUNAWAY_TREND_LAG_S` 2.0 $\rightarrow$ **10.0 s**: **2.8 $\times$** the longest beat, **5.6–13 $\times$** shorter than τ_{env} .

The remaining 14.4% is not thrash: $7 \times 5 \text{ s}$ of mandatory hold = 35 s = 14.3% of 244.5 s.

% of time	DAMP	FAULT	CALIB	entries
LAG = 2.0	54.1	25.2	20.7	10
LAG = 10.0	77.5	14.4	8.2	7

Two hardware runs, 244.5 s, five hand kicks, back to back, same gains.

The fix is not free. Run the simulated suite at each lag with everything else identical and **exactly one assertion flips**, the wrong-sign detector: with ch2's gain inverted over 120 s, a 2 s lag trips the breaker ≥ 3 times and a 10 s lag trips it **2**, against a check demanding 3. **The wrong sign is still caught:** the companion faults > 0 passes at both. A breaker that no longer reads a 1.55 s beat as growth also fires less often on something genuinely growing.

77.5% is not a win over beta's 83.1%; different kick schedule.

Defect 6: stationarity is not quietness

- 1. What it was supposed to do.** Measure the room's noise floor during CALIBRATING, when the gain is zero and nothing is being driven.
- 2. What it did.** Gain off is not the same as quiet. A calibration taken while the optic is still ringing measures *the ringing*, and every threshold in the run is then a ratio against that inflated number.
- 3. The evidence.** Caught on hardware: a fresh calibration produced a floor **11.5×** the trusted one. Taking alpha's ch0 floor as the trusted value:

	trusted	inflated 11.5×
baseline (V)	0.1804	2.075
runaway line, 1.8×	0.325	3.735
lock line, 0.35×	0.063	0.726

The breaker goes blind and the rig would announce LOCKED at 11.5× the residual motion it should demand. And persist-baseline would carry that across restarts.

- 4. The fix, and the number that moved.**
baseline-sanity: a fresh floor that fails the check is **refused** and the trusted one kept. **On the bench the 11.5×** floor was refused.

The guard needs a guard.

MAX_BASELINE_REFUSALS = 3 stops it refusing forever if the room genuinely *did* get louder.

What the persistence buys. The noise floor is a property of the room, not of the run, so delta writes it to disk and reuses it across restarts, behind a config fingerprint, an age limit and this sanity check.

cold-start calibration	before	after
	20 s	2.0 s

This finishes runaway-baseline, nine versions later. alpha made one bad sub-window outvotable; delta makes a bad *whole calibration* refusable.

Open defect 1: one undriven channel vetoes the announcement

1. What it was supposed to do. Print `*** LOCKED ***` once every channel that is supposed to be under control has sat below $0.35 \times$ its baseline for 5s. **The lock time is the project's stated deliverable.**

2. What it did. The test is `self.locked[live].all()` with `live = enabled & healthy`, and `ENABLE_CHANNEL` is `True` on *all eight*. A **quorum** is the set of channels that must all agree before the rig will say anything, and this one includes a channel nothing is driving.

3. The evidence. Zero `*** LOCKED ***` lines in any run of the 2026-08-06 session, delta included, yet the rig does lock: the four `ch{i}_locked` columns are simultaneously 1 for **6 distinct episodes** on the good delta run. **One undriven channel vetoes the whole rig.** A reporting bug, not a damping failure.

	in live?	driven?	ratio	locked
a0	yes	yes	< 0.35	1
a1	yes	yes	< 0.35	1
a2	yes	yes	< 0.35	1
a3	yes	yes	< 0.35	1
a5	yes	no	1.26–2.53	0
a4	no (NOSIG)	no	n/a	n/a
a6	no (NOSIG)	no	n/a	n/a
a7	no (NOSIG)	no	n/a	n/a

a4, a6 and a7 demote themselves to NOSIG and leave live. **a5 does not:** it stays healthy, so it stays in the quorum, but `STEADY_GAIN[5] = 0` so nothing drives it.

4. Not never any more, 2026-08-07. `epsilon` announced `LOCKED` on hardware **for the first time in the project:** 12.1 s after gain was applied, **4 of the 8 channels driven. The quorum is still unfixed:** dropping zero-gain channels from it is one line, not yet applied, so the eight-channel configuration above still cannot announce.

Open defect 2: no OSEM rests at mid-scale

1. **What it was supposed to do.** Each flag should sit in the middle of its sensor's range, so the optic has equal room to swing either way before the reading hits a rail. The ADC reads **0 to 1023 counts**, so mid-scale is **511.5**.

2. **What it did. Every channel rests above mid-scale**, so every channel meets its *top* rail first, and a railed sensor is outside its linear region. The velocity estimate is therefore wrong exactly when the optic is moving hardest and the loop is pushing hardest.

3. **The evidence.** Resting counts, CALIBRATING, 22 246 samples, outputs at bias, 2026-08-06:

	ch0	ch1	ch2	ch3
resting count	601.2	630.3	686.5	681.8
room above	422	393	336	341
room below	601	630	686	682

ch2 is worst: **2.0×** as much room below as above.

of 251 280 samples	ch0	ch1	ch2	ch3
railed	2.9%	0.8%	3.6%	0.9%
rail interlock caught	136	0	51	0

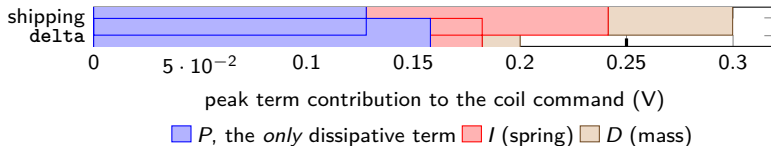
The interlock needs the pin *sustained*, and this clipping is transient, so it caught **187** of **20 696**.

About 99% of the clipping is invisible to every interlock.

4. **Open. No fix applied.** The experiment costs no code and has **not been run**, so there is no result to report. Coil bias is a static force: step one channel 0.10 → 0.40 V at zero gain and log the resting counts. **If they move**, the offset is electromagnetic and a per-channel bias trim removes it at source; **if not**, it is mechanical flag alignment and no tuning helps.

Open defect 3: the clipping is arithmetic, not bad luck

1. **What it was supposed to do.** The three terms together should fit inside the coil drive's half-window, $V_{MAX} - BIAS = 0.250\text{ V}$, so the loop never asks for a push the hardware cannot deliver.
2. **What it did.** The shipping gains peak at $0.1279 + 0.1135 + 0.0584 = \mathbf{0.2998\text{ V}}$, **120% of budget**. The channels clip because the demand exceeds the window, not because of bad luck.



Dashed line: the actuator half-window, $V_{MAX} - BIAS = 0.25\text{ V}$.

3. **The evidence.** The largest single item is K_i at **0.1135 V, 38% of the peak**, and it dissipates *nothing*: the integrator sits behind the 0.4 Hz highpass, so $i = -K_i \text{ bp}$, and bp carries no DC for an integrator to reject. **Nothing in this project justifies K_i or K_d** : no scope trace, no sweep, no sentence, where $|K_p| = 0.030$ has one. Switching them on moved the ch0 amplitude ratio from **0.025 to 0.028**.

4. **Open. No fix applied.** delta cuts K_i and K_d to 15% and raises K_p to the 0.035 ceiling. The numbers below are computed from the measured per-term peaks, **not confirmed on the bench**.

	shipping	delta
peak demand	0.2998 V	0.2001 V
authority that <i>damps</i>	43%	88%

Four gains, two observable degrees of freedom

The data. Fit the plant from the dither run and ask what the added damping depends on:

free gains per term, ch0–ch3	4
observable degrees of freedom	2
undetermined dimension	2

Damping constrains exactly **two linear functionals** of a four-element gain vector.

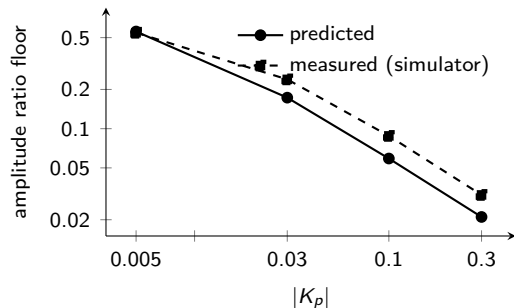
So a two-dimensional family of gain vectors damps identically. Every direction inside it is indistinguishable by measurement, and the optimiser picks its point **by minimum actuator effort, not by measurement.**

This is ill-posedness, not a missing experiment. More dither does not shrink a null space.

Flat is as good as any point in that family. At the 0.035 ceiling, flat ± 0.035 reaches **90% of the achievable maximum**, worst-mode added decay 0.0739 s^{-1} .

The measurable lever is the actuator budget, not the gain shape.

The floor: why more gain is the only way down



The ratio settles at **0.23**, and that is the *steady state of a continuously forced oscillator*, not a failure to converge. Remove the drive mid-run and it decays with $\tau = 3.4$ s damped against 15.9 s gain-off (predicted 2.76 / 15.92).

The floor tracks $\frac{\text{intrinsic}}{\text{intrinsic} + \text{added}}$ damping, so it falls monotonically with gain.

The floor falls with gain, and gain is capped at $|K_p| = 0.035$ by the instability bound.

Those two facts are the vice the project sits in.

The residual **0.0089 V** is the actuator's 0.5 mV deadband and 10 ms throttle, *not* sensing: 0.0089/0.0087/0.0094 V at 20/5/0 mV of injected sensor noise. Fixing it moves 0.009 to 0.009 while you sit at 0.23.

Still open, in the order I would do them

1. **Eight channels still cannot announce LOCKED.** a5 vetoes the quorum at ratio 1.26–2.53 against 0.35 while nothing drives it. **One line.** (epsilon did lock on four driven channels, 2026-08-07.)
2. **The bias sweep.** Step one channel $0.10 \rightarrow 0.40$ V at zero gain, log resting counts. Settles whether the 601/630/687/682 offset is electromagnetic or mechanical. **No code.**
3. **Confirm delta's gain cut on the bench.** Peak demand $0.2998 \rightarrow 0.2001$ V, dissipative share $43\% \rightarrow 88\%$.
4. **Cut LOCK_SUSTAIN_S = 5.0**, on its own: five of the ten-second budget confirming a lock that already happened.
5. **Gate fault re-engagement on amplitude**, not a fixed 5 s, which is 35 of delta's 35.2 fault-seconds.
6. **Kalman velocity estimator**, unblocked: f to 0.0001 Hz, $\tau > 138$ s so Q stops mattering, measurement noise measured.
7. **Read the make check failures**; only two have ever been looked at, and the count moves between runs.
8. **a4, a6, a7.** Connected, near mid-scale, $\sim 1/100$ the in-band gain of a0–a3. Cause not established.

Learn the model, the estimator and the anomaly detector; keep the control law and the interlocks classical and certifiable.